

BudgetWise AI - Expense Forecaster

1. Project Overview

1.1 Executive Summary

BudgetWise AI is an intelligent expense forecasting and personal financial management platform that leverages artificial intelligence to help individuals and businesses gain control over their finances. The system combines automated transaction categorization, advanced forecasting algorithms, and goal-oriented financial planning to provide actionable insights and predictions.

1.2 Key Value Propositions

- **AI-Powered Categorization:** Automatic transaction classification using NLP and machine learning
- **Intelligent Forecasting:** Advanced time-series prediction using Prophet and statistical models
- **Goal Management:** AI-driven financial goal setting and achievement coaching
- **Real-time Analytics:** Comprehensive spending analysis and visualization

1.3 Technical Achievement

- The project successfully implements a full-stack AI financial platform with:
- **Backend:** Python Flask REST API with SQLite database
- **Frontend:** Streamlit-based interactive dashboard
- **AI/ML:** Integration of Prophet, NLTK, and custom forecasting algorithms
- **Security:** JWT-based authentication and data encryption
- **Deployment:** Container-ready architecture

2. System Requirements & Problem Statement

2.1 Problem Statement

Current Challenges in Personal Finance Management: Manual expense tracking is time-consuming and error-prone Lack of intelligent categorization leads to poor spending insights Difficulty in predicting future expenses and cash flow No AI-driven guidance for financial goal achievement Limited real-time budget monitoring and alerts

2.2 Solution Objectives

Automated Processing: AI-powered transaction categorization Predictive Analytics: Advanced expense forecasting Goal-Oriented Planning: AI coaching for financial targets Real-time Monitoring: Live dashboards and alerts Multi-user Support: Individual and administrative views

2.3 Functional Requirements

Module	Requirements
User Management	Registration, login, profile management
Transaction Handling	Manual entry, CSV upload, categorization
Forecasting	Time-series prediction, trend analysis
Goals Management	Goal creation, progress tracking, AI coaching
Reporting	Category analysis, monthly summaries, insights

2.4 Non-Functional Requirements

- Performance: Sub-second response for most operations
- Security: JWT authentication, data encryption
- Scalability: Support for multiple concurrent users
- Usability: Intuitive interface with minimal learning curve
- Reliability: 99% uptime with proper error handling

3.Technology Stack

3.1 Backend Technologies

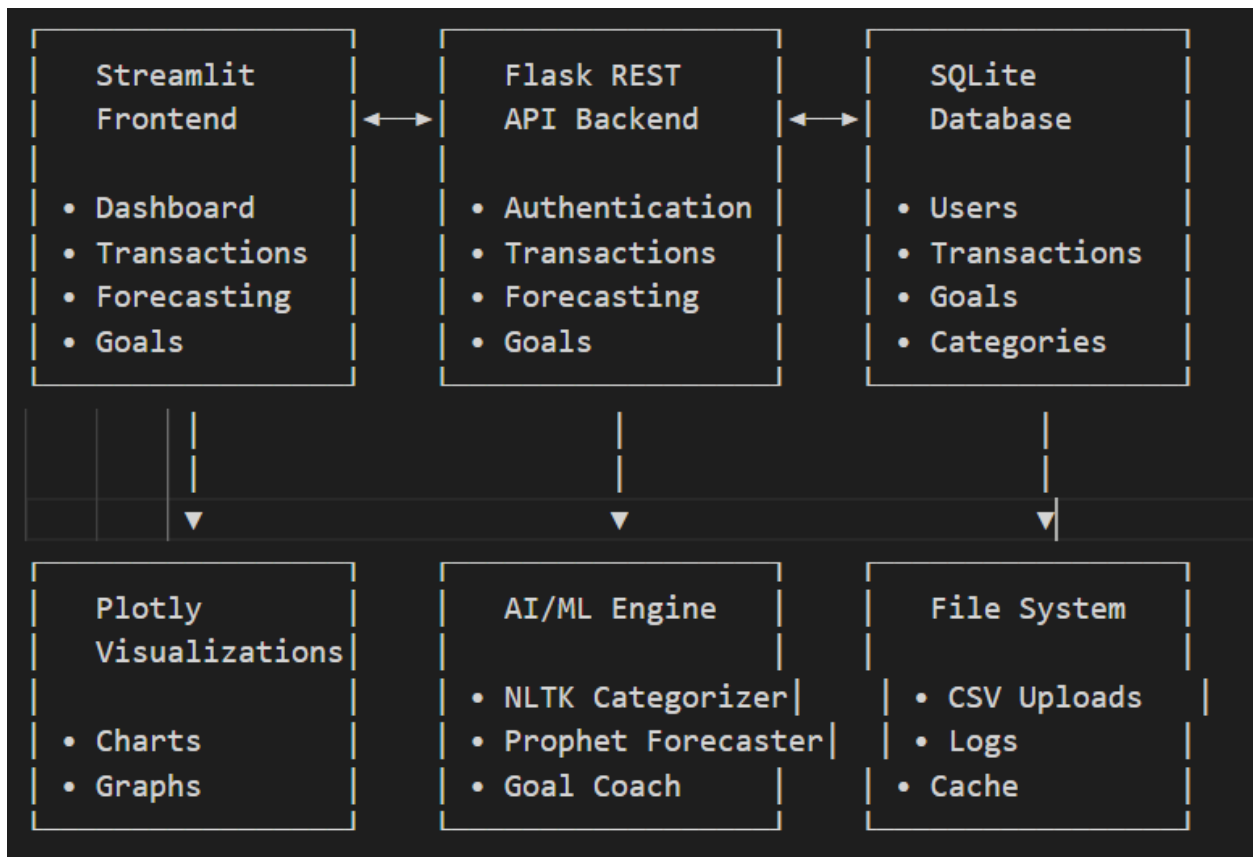
Component	Technology	Purpose
Framework	Python Flask	REST API development
Database	SQLite	Data persistence
Authentication	JWT	Secure user authentication
ML Framework	Prophet, scikit-learn	Forecasting and categorization
NLP Library	NLTK	Transaction description processing
Data Processing	Pandas, NumPy	Financial data analysis

3.2 Frontend Technologies

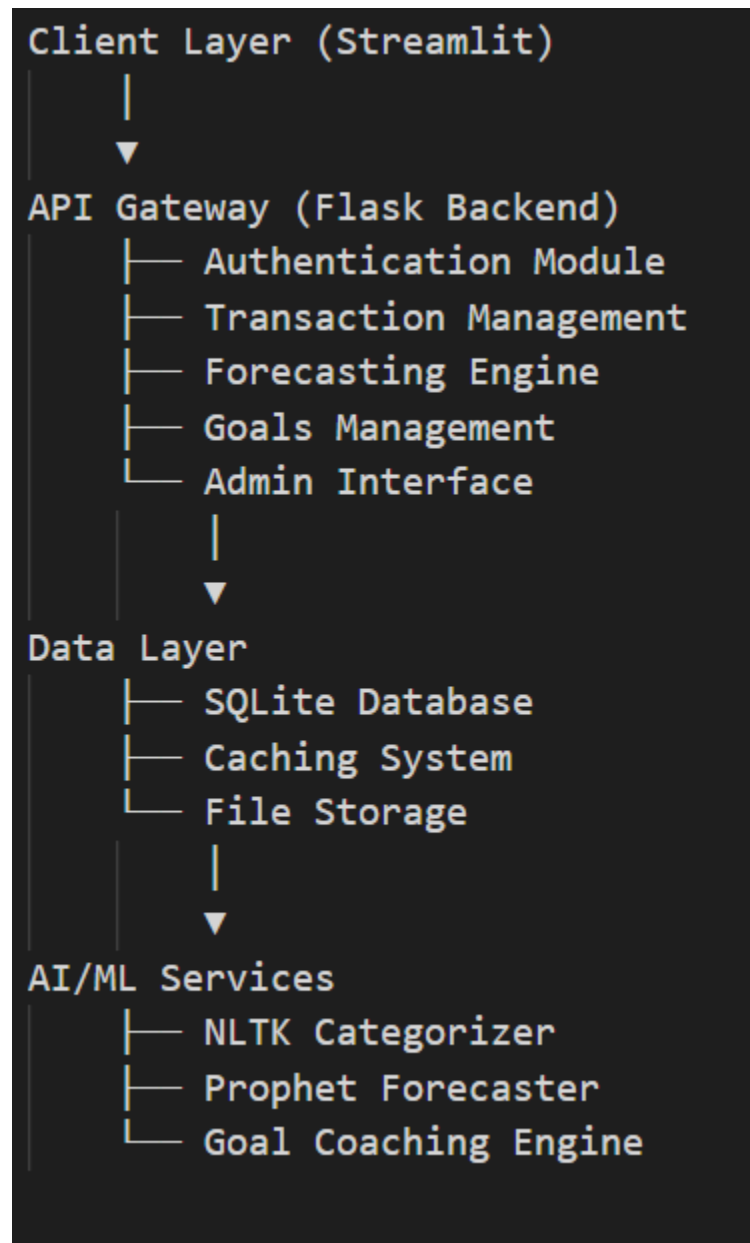
Component	Technology	Purpose
Framework	Streamlit	Web application interface
Visualization	Plotly, Matplotlib	Charts and graphs
UI Components	Streamlit native	Forms, tables, navigation
Styling	Custom CSS	Interface theming

4. System Architecture

4.1 High-Level Architecture Diagram



4.2 Component Architecture



5.Database Design

5.1 Database Schema

Table: users			
Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY	Unique user identifier
username	TEXT	NULLABLE	User display name
email	TEXT	UNIQUE, NOT NULL	User email for login
password_hash	TEXT	NOT NULL	Encrypted password
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date
Table: transactions			
Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY	Transaction identifier
user_id	INTEGER	FOREIGN KEY	Associated user
date	DATE	NOT NULL	Transaction date
amount	DECIMAL(10,2)	NOT NULL	Transaction amount
description	TEXT	NULLABLE	Transaction description
category	TEXT	NULLABLE	AI-categorized spending category
type	TEXT	CHECK(income/expense)	Transaction type
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Table: financial_goals			
Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY	Goal identifier
user_id	INTEGER	FOREIGN KEY	Goal owner
goal_name	TEXT	NOT NULL	Goal description
goal_type	TEXT	CHECK(savings/spending_reduction/category_budget)	Goal category
target_amount	DECIMAL(10,2)	NOT NULL	Target amount
current_amount	DECIMAL(10,2)	DEFAULT 0	Current progress
target_date	DATE	NOT NULL	Goal deadline
category	TEXT	NULLABLE	Associated spending category
description	TEXT	NULLABLE	Goal details
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Last update timestamp

Table: goal_savings			
Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY	Savings record ID
goal_id	INTEGER	FOREIGN KEY	Associated goal
amount	DECIMAL(10,2)	NOT NULL	Savings amount
saved_date	DATE	NOT NULL	Savings date
description	TEXT	NULLABLE	Savings note
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record timestamp

5.3 Indexes and Optimization

Performance indexes

```
CREATE INDEX idx_transactions_user_id ON transactions(user_id); CREATE INDEX  
idx_transactions_date ON transactions(date); CREATE INDEX idx_financial_goals_user_id  
ON financial_goals(user_id); CREATE INDEX idx_goal_savings_goal_id ON  
goal_savings(goal_id);
```

Auto-update timestamp trigger

```
CREATE TRIGGER update_financial_goals_timestamp AFTER UPDATE ON financial_goals  
FOR EACH ROW BEGIN UPDATE financial_goals SET updated_at = CURRENT_TIMESTAMP  
WHERE id = NEW.id; END;
```

6.Module Specifications

6.1 Authentication Module

Purpose: Secure user registration, login, and session management

Key Features:

JWT-based authentication Password hashing with Werkzeug security Session management
with token expiration Admin user detection and privileges

Endpoints:

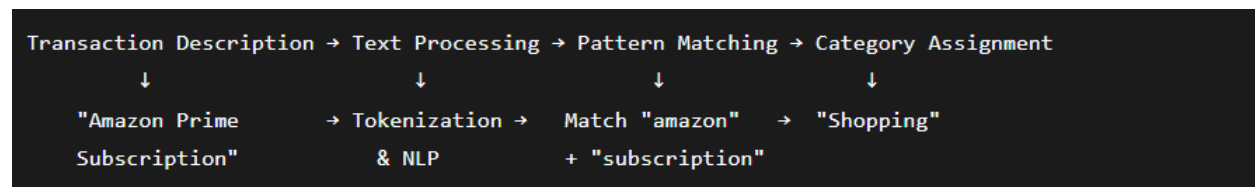
POST /auth/register - User registration POST /auth/login - User authentication JWT-
protected routes for all subsequent requests

6.2 Transaction Management Module

Purpose: Handle all financial transaction operations

Key Features:

Manual transaction entry with AI categorization Bulk CSV upload with automated processing Transaction categorization using NLP Category override and manual correction Transaction history with filtering and sorting AI Categorization Process:



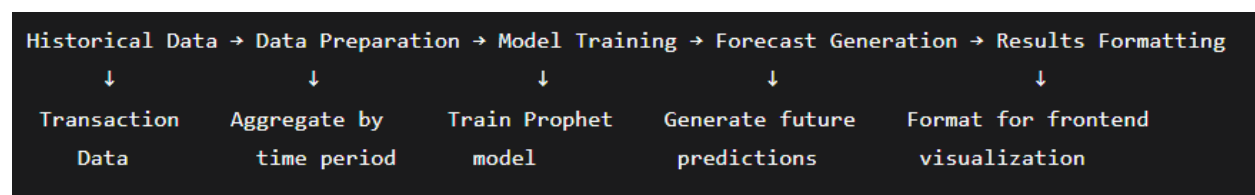
6.3 AI Forecasting Module

Purpose: Generate expense predictions using advanced algorithms

Key Components:

Prophet Integration: Facebook's forecasting library for time-series data Simple Fallback Model: Moving average-based forecasting when Prophet unavailable Anomaly Detection: Z-score based outlier identification Seasonality Analysis: Weekly, monthly, and yearly pattern recognition

Forecasting Process:



6.4 Goals Management Module

Purpose: Financial goal setting, tracking, and AI coaching

Key Features:

Goal creation with target amounts and deadlines Progress tracking with visual indicators Savings contribution system AI-powered goal coaching and feasibility analysis Goal analytics and completion metrics

AI Coaching Analysis:

Financial capacity assessment Success probability calculation Actionable recommendation generation Timeline feasibility analysis

6.5 Administration Module

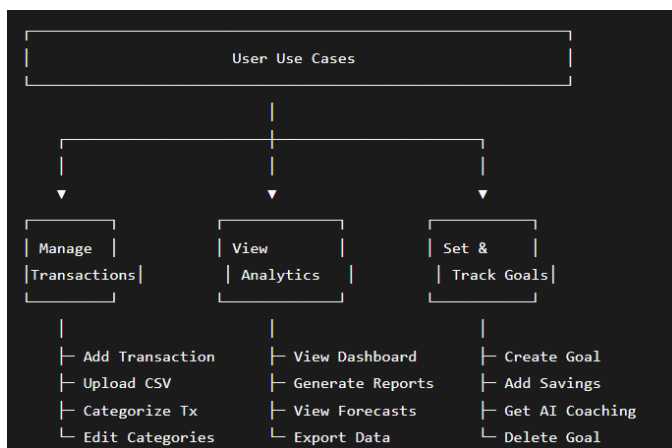
Purpose: System management and user oversight

Key Features:

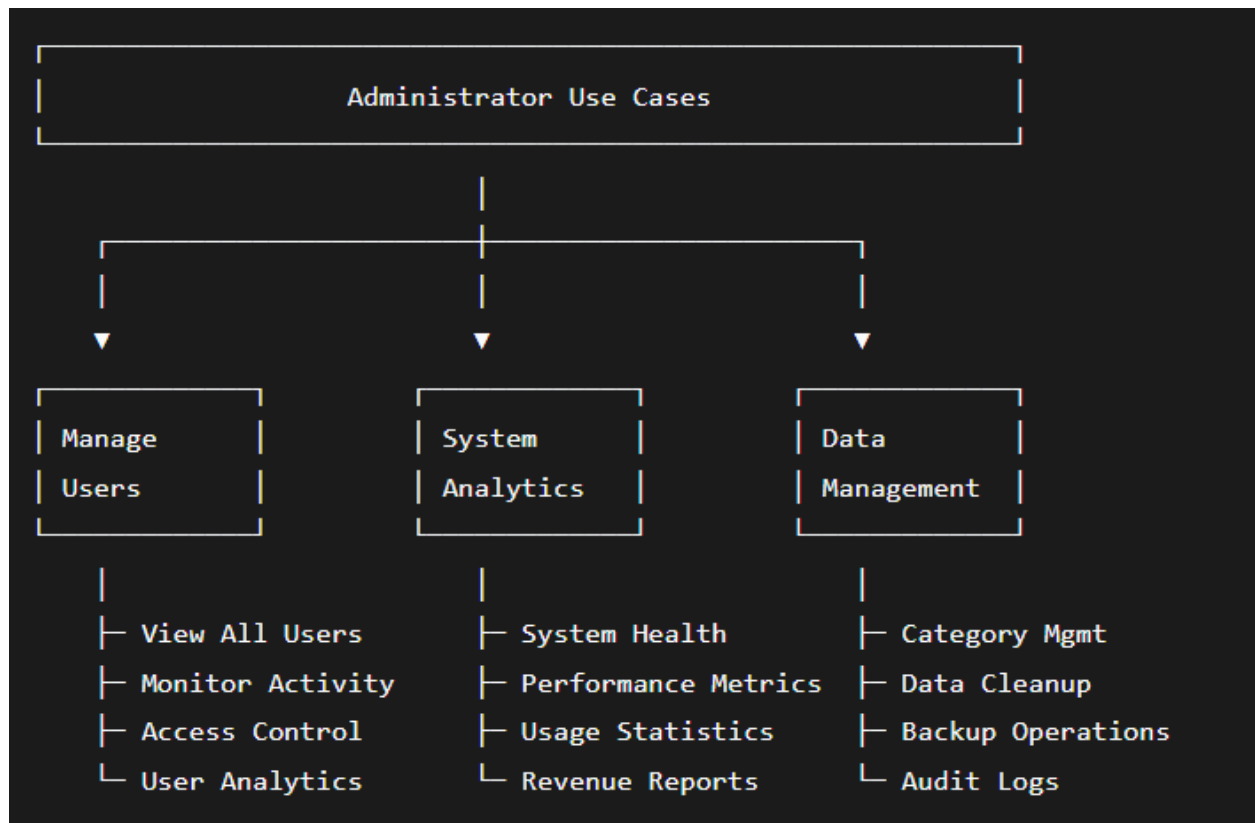
User management and activity monitoring System analytics and health checks Category management and normalization Data cleanup and maintenance operations Transaction oversight across all user

7. Use Case Diagrams

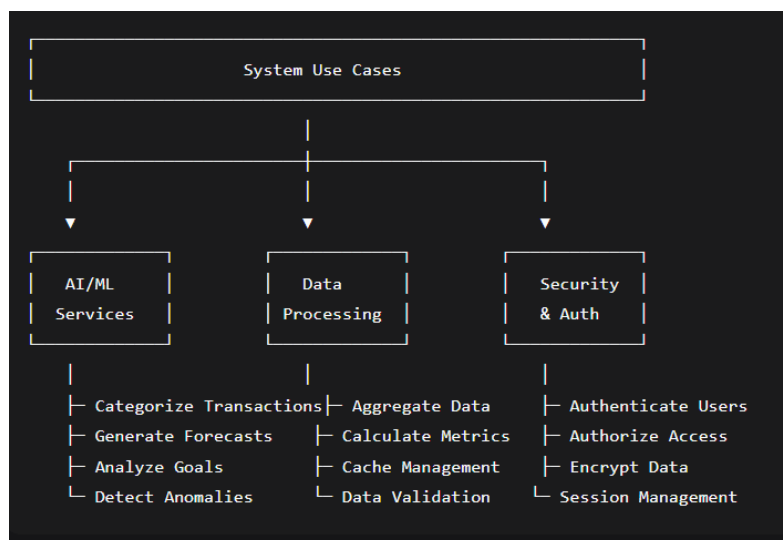
7.1 User Use Case Diagram



7.2 Administrator Use Case Diagram

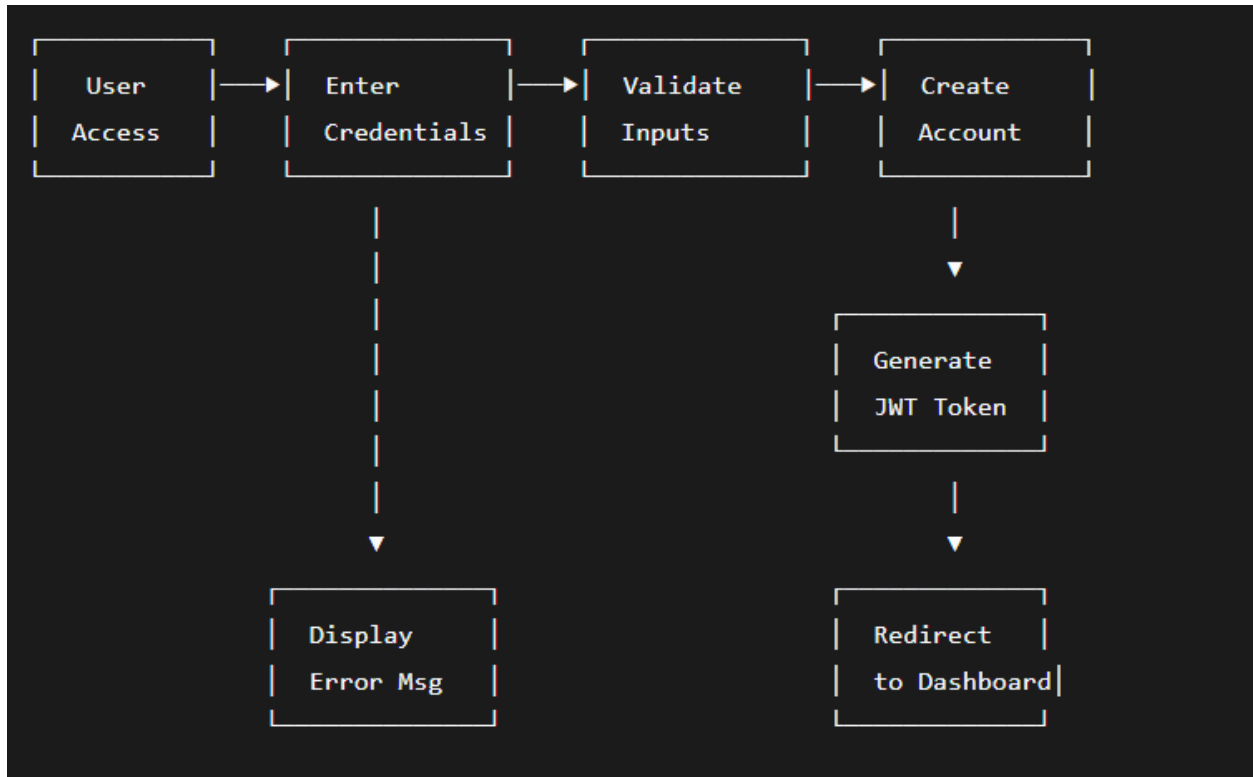


7.3 System Use Case Diagram

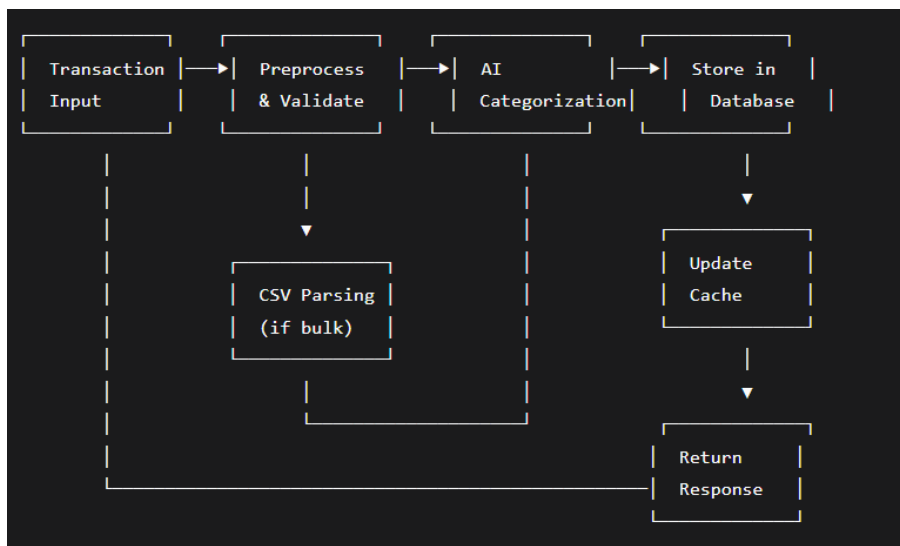


8. Workflow Diagrams

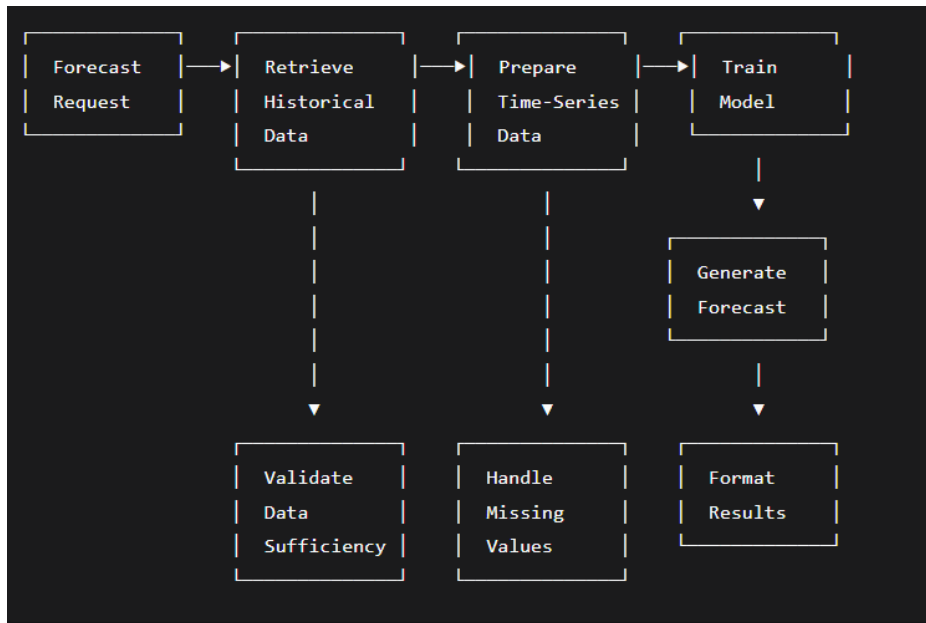
8.1 User Registration & Login Workflow



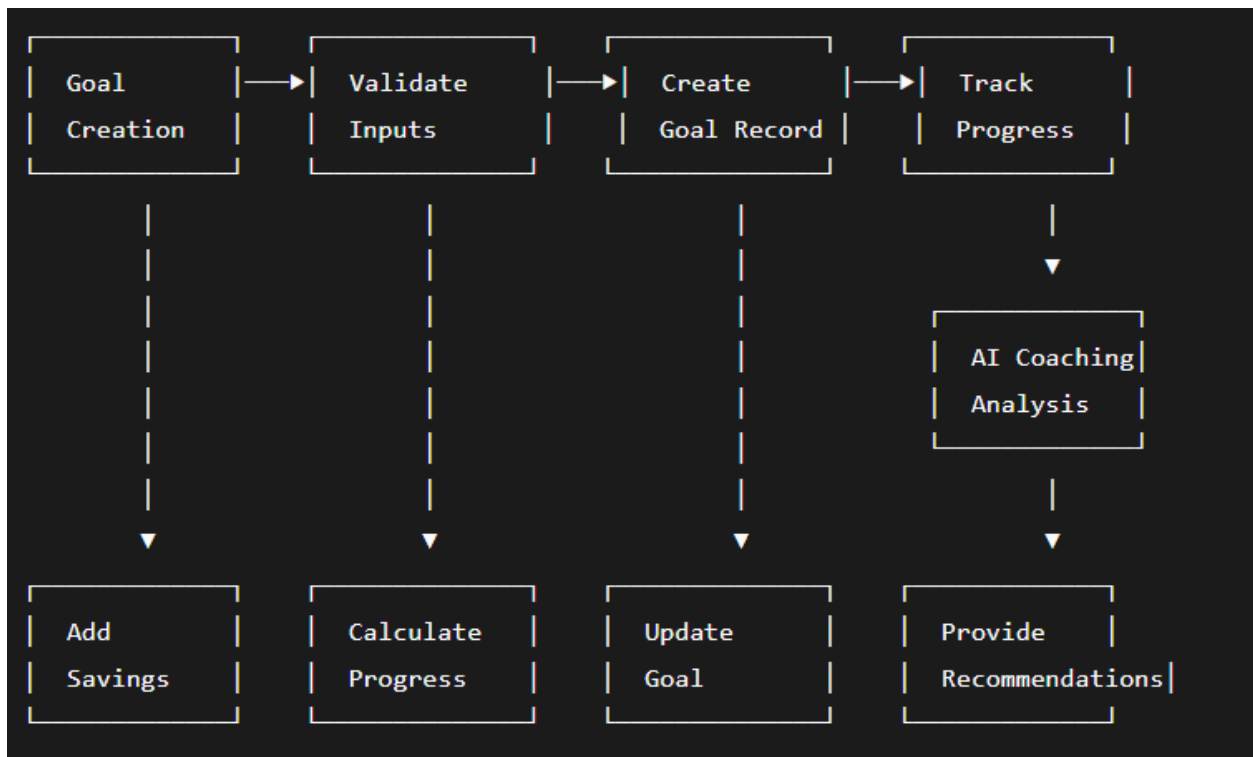
8.2 Transaction Processing Workflow



8.3 Forecasting Workflow



8.4 Goal Management Workflow



9. API Endpoints Documentation

9.1 Authentication Endpoints

Method	Endpoint	Purpose	Parameters
POST	/auth/register	User registration	email , password
POST	/auth/login	User authentication	email , password

9.2 Transaction Endpoints

Method	Endpoint	Purpose	Parameters
POST	/transactions	Add single transaction	date , amount , description , type , category
POST	/transactions/bulk	Bulk CSV upload	file (CSV)
GET	/transactions	List user transactions	limit , offset
PUT	/transactions/{id}/category	Update category	category

9.3 Reporting Endpoints

Method	Endpoint	Purpose	Parameters
GET	/reports/category	Category spending	days (time period)
GET	/reports/monthly	Monthly summaries	months (period count)
GET	/reports/overview	Financial overview	None

9.4 Forecasting Endpoints

Method	Endpoint	Purpose	Parameters
POST	<code>/forecast</code>	Generate forecast	<code>category</code> , <code>months_ahead</code> , <code>frequency</code>
GET	<code>/forecast/categories</code>	Available categories	None
POST	<code>/forecast/compare</code>	Multiple forecasts	<code>categories</code> , <code>months_ahead</code>
GET	<code>/forecast/performance</code>	Model metrics	None

9.5 Goals Endpoints

Method	Endpoint	Purpose	Parameters
POST	<code>/goals</code>	Create goal	<code>goal_name</code> , <code>goal_type</code> , <code>target_amount</code> , <code>target_date</code>
GET	<code>/goals</code>	List user goals	None
DELETE	<code>/goals/{id}</code>	Delete goal	None
POST	<code>/goals/{id}/savings</code>	Add savings	<code>amount</code> , <code>description</code>
GET	<code>/goals/{id}/coaching</code>	AI coaching	None
GET	<code>/goals/analytics</code>	Goal analytics	None

9.6 Administration Endpoints

Method	Endpoint	Purpose	Parameters
GET	<code>/admin/check-access</code>	Verify admin rights	None
GET	<code>/admin/users</code>	List all users	None
GET	<code>/admin/transactions</code>	All transactions	<code>limit</code>
GET	<code>/admin/analytics</code>	System analytics	None
GET	<code>/admin/categories</code>	Category statistics	None
POST	<code>/admin/categories/update</code>	Update categories	<code>old_category</code> , <code>new_category</code>
GET	<code>/admin/system/health</code>	System health	None
POST	<code>/admin/system/cleanup</code>	Data cleanup	<code>type</code>

10.AI/ML Components Specification

10.1 NLTK Transaction Categorizer Architecture:

- Hybrid rule-based + machine learning approach
- Processing Pipeline:
 - Text Preprocessing
 - Tokenization and stop-word removal
 - Lemmatization and stemming
 - Part-of-speech tagging
- Rule-Based Categorization
 - Exact pattern matching
 - Fuzzy matching with similarity thresholds
 - Brand-specific context rules
 - Category priority weighting
- Machine Learning Fallback
 - TF-IDF vectorization
 - Cosine similarity matching
- Training data with comprehensive examples
- Category Taxonomy:
 - Essential: Rent, Utilities, Healthcare, Insurance, Loan_Repayment
 - Discretionary: Entertainment, Dining, Shopping, Travel
 - Flexible: Groceries, Transport, Education
 - Income: Salary, Bonus
 - Miscellaneous: Uncategorized transactions

10.2 Prophet Forecasting Engine Model Configuration:

- Seasonality: Weekly, monthly, and yearly patterns
- Holiday Effects: Country-specific holiday integration
- Changeoint Detection: Automatic trend change identification
- Uncertainty Intervals: 80% confidence intervals
- Fallback Mechanism:
 - Simple moving average model when Prophet unavailable
- Trend analysis and seasonal adjustment
- Confidence interval estimation
- Performance Metrics:
 - Mean Absolute Percentage Error (MAPE)
 - Root Mean Square Error (RMSE)
- Anomaly detection accuracy

10.3 Goal Coaching AI Analysis Components:

- Financial Capacity Assessment: Income vs. expense analysis
- Savings Feasibility: Realistic savings capacity calculation
- Timeline Analysis: Goal achievement probability
- Spending Optimization: Category-specific reduction opportunities
- Coaching Output:
 - Success probability percentage
 - Actionable recommendations
 - Monthly savings targets
 - Timeline adjustments

11.Security Implementation

11.1 Authentication Security JWT Tokens: Stateless authentication with expiration

- Password Hashing: Werkzeug security with salt
- Input Validation: Comprehensive request validation
- CORS Protection: Configured cross-origin resource sharing

11.2 Data Security SQL Injection Prevention: Parameterized queries

- XSS Protection: Input sanitization and output encoding
- Data Encryption: Sensitive data encryption at rest
- Access Control: User-level data isolation

11.3 Administrative Security Admin Detection: Hardcoded admin user validation

- Privilege Separation: Distinct user and admin functionalities
- Audit Logging: System operation tracking
- Data Isolation: User data access restrictions

12. Performance Optimization

12.1 Caching Strategy Frontend Caching: Transaction data with timestamp validation

- Database Indexing: Strategic indexes on frequently queried columns
- Query Optimization: Efficient database query patterns
- Connection Pooling: Database connection reuse

12.2 Response Time Targets Dashboard Load: < 2 seconds

- Transaction List: < 1 second
- Forecast Generation: < 10 seconds
- Goal Coaching: < 5 seconds

12.3 Scalability Considerations Stateless API: Enables horizontal scaling

- Database Optimization: Efficient query patterns
- Resource Management: Connection and memory management
- Error Handling: Graceful degradation under load

13. Error Handling & Validation

13.1 Input Validation Data Type Checking: Type conversion and validation

- Range Validation: Amount and date boundary checks
- Format Validation: Date format and email validation
- Business Logic: Goal feasibility and transaction validity

13.2 Error Response Standardization

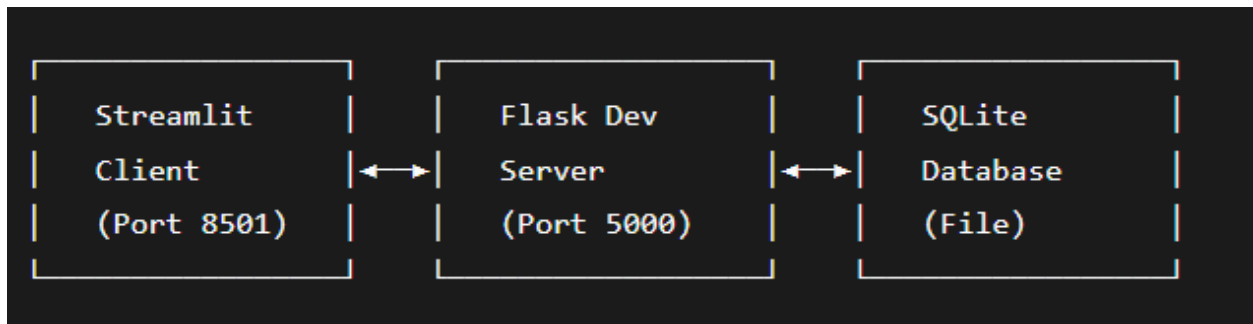
```
{  
  
  "success": false,  
  
  "error": "Descriptive error message",  
  
  "code": "ERROR_CODE",  
  
  "details": {}  
}
```

13.3 Exception Categories Authentication Errors: Invalid credentials, expired tokens

- Validation Errors: Invalid input data, missing required fields
- Database Errors: Constraint violations, connection issues
- AI/ML Errors: Model training failures, data insufficiency
- System Errors: Internal server errors, external service failures

14. Deployment Architecture

14.1 Development Environment



14.2 Production Readiness Containerization: Docker support for easy deployment

- Environment Configuration: Environment variable-based configuration
- Logging: Comprehensive application logging
- Health Checks: System health monitoring endpoints

14.3 Deployment Considerations Database Migration: Schema version management

- Data Backup: Regular database backup procedures
- Security Hardening: Production security configurations
- Monitoring: Application performance monitoring

15. Testing Strategy

15.1 Test Categories Unit Testing: Individual component testing

- Integration Testing: Module interaction testing
- API Testing: Endpoint functionality validation
- UI Testing: Frontend interface testing
- Performance Testing: Load and stress testing

15.2 Test Coverage Areas User authentication and authorization

- Transaction processing and categorization
- Forecasting algorithm accuracy
- Goal management functionality
- Administrative operations
- Error handling and edge cases

16. Compliance & Data Management

16.1 Data Privacy User Data Isolation: Strict separation between user data

- Minimal Data Collection: Only essential user information
- Data Encryption: Sensitive data protection
- Access Logging: User activity tracking

16.2 Financial Data Handling Data Accuracy: Validation and verification processes

- Audit Trail: Transaction history preservation
- Data Integrity: Constraint enforcement and validation
- Backup Procedures: Regular data backup schedules

17. Maintenance & Operations

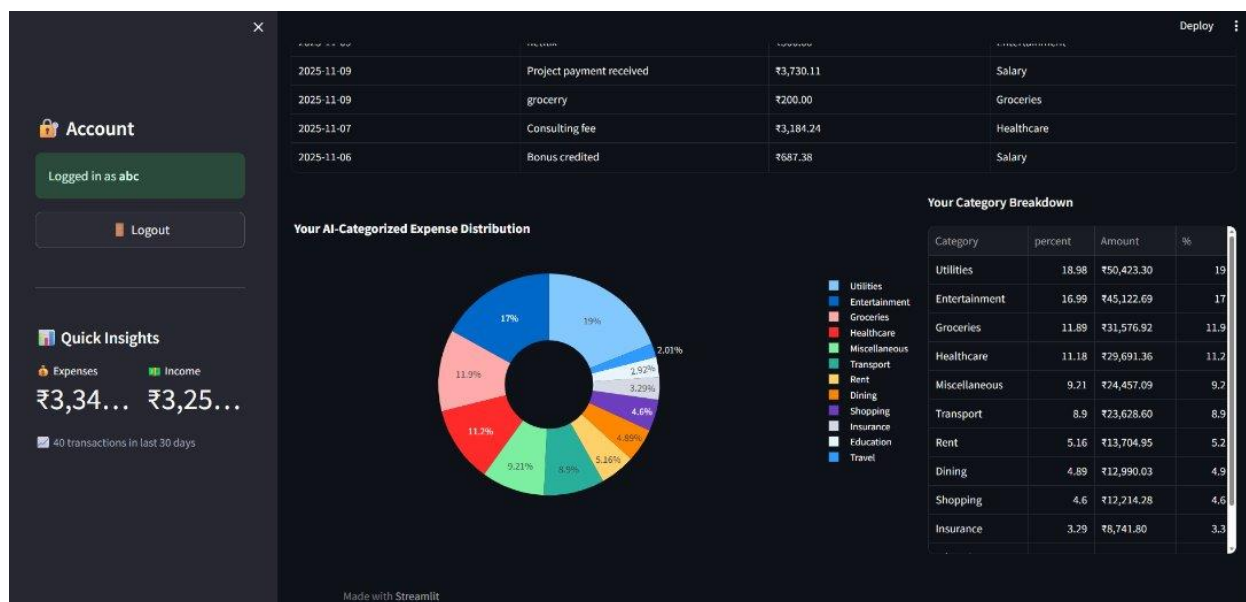
17.1 Regular Maintenance Tasks Database Cleanup: Orphaned data removal

- Cache Management: Cache invalidation and refresh
- Log Rotation: Log file management
- Performance Monitoring: System performance tracking

17.2 Administrative Operations User Management: User account administration

- Category Management: Transaction category normalization
- System Monitoring: Health and performance monitoring
- Data Analytics: Usage pattern analysis

18. Screenshots



Account

Logged in as abc

Logout

Quick Insights

ExpensesIncome

₹3,34... ₹3,25...

40 transactions in last 30 days

Date

2025/11/10

Type

expense

Amount

300.00

Description

netflix

Choose category (AI auto-categorizes if empty):

Add Transaction

Choose CSV file

Drag and drop file here

Limit 200MB per file • CSV

Browse files

Your Transactions

Time Period

Last 30 days

Search descriptions

Sort by

Date (newest)

amount	category	date	description	id	type
₹300.00	Entertainment	2025-11-10	netflix	3,003	expense
₹1,000.00	Healthcare	2025-11-10	gym	3,005	expense
₹300.00	Entertainment	2025-11-10	netflix	3,004	expense
₹300.00	Entertainment	2025-11-09	netflix	1	expense

Account

Logged in as abc

Logout

Quick Insights

ExpensesIncome

₹3,34... ₹3,25...

40 transactions in last 30 days

value

120k

100k

80k

60k

40k

20k

0

Aug 10 2025

Aug 24

Sep 7

Sep 21

Oct 5

Oct 19

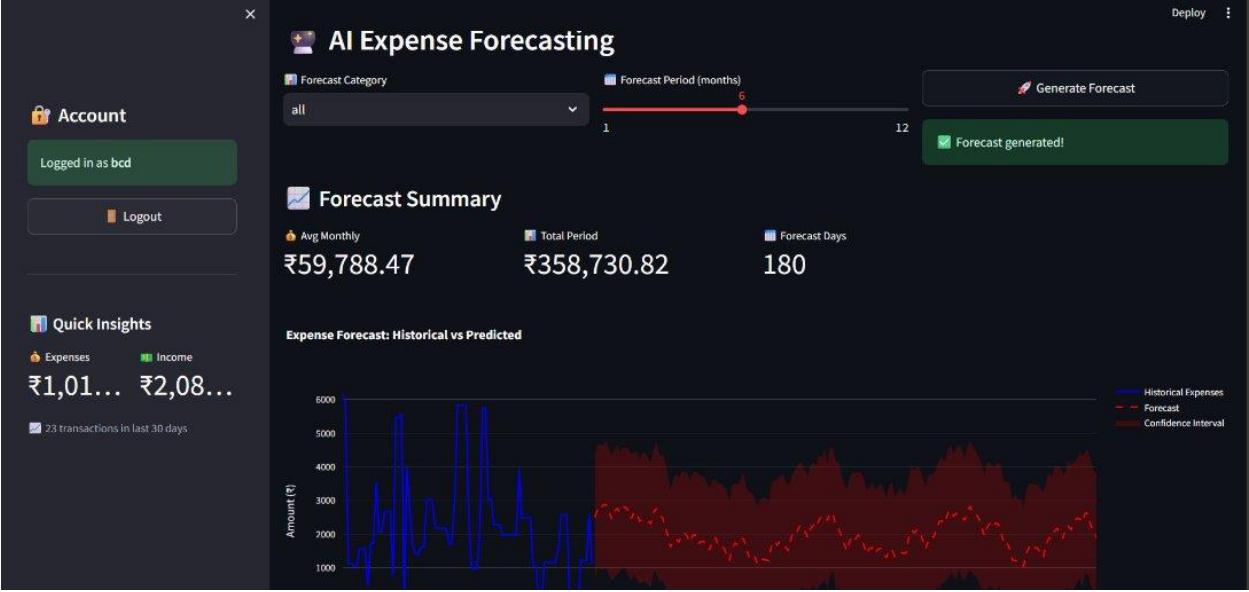
month

expense

Category Report

Spending by Category

category	percent	Amount	%
Utilities	18.98	₹50,423.30	19
Entertainment	16.99	₹45,122.69	17
Groceries	11.89	₹31,576.92	11.9
Healthcare	11.18	₹29,691.36	11.2
Miscellaneous	9.21	₹24,457.09	9.2
Transport	8.9	₹23,628.60	8.9
Rent	5.16	₹13,704.95	5.2
Dining	4.69	₹12,090.03	4.6
Shopping			



BudgetWise Admin Dashboard

System Administration & Analytics

Restricted Access - Administrators Only

Admin Emailyour-admin@email.comPassword

Login as Administrator





Deploy

🔧

BudgetWise Admin Dashboard

Logged in as: [admin@budgetwise.com](#)

Dashboard

User Management

Transactions

Category Management

System Tools

User Management

User Overview (3 Users)

ID	Email	Joined	Transactions	Last Activity
3	bcd	2025-11-15	1,000	2025-11-09
2	abc	2025-11-15	1,000	2025-11-09
1	admin	2025-11-15	0	Never

User Statistics

Active Users

2

Avg Transactions

666.7

Newest User

bcd...

Deploy

50

500

Recent Transactions (100 loaded)

ID	User	Date	Amount	Type	Category	Description
2,000	bcd	2025-11-09	₹3,730.11	income	Salary	Project payment received
1,000	abc	2025-11-09	₹3,730.11	income	Salary	Project payment received
1,999	bcd	2025-11-07	₹3,184.24	income	Healthcare	Consulting fee
999	abc	2025-11-07	₹3,184.24	income	Healthcare	Consulting fee
1,998	bcd	2025-11-06	₹687.38	income	Salary	Bonus credited
998	abc	2025-11-06	₹687.38	income	Salary	Bonus credited
1,997	bcd	2025-11-05	₹1,099.97	expense	Groceries	Office supplies - stationery
1,996	bcd	2025-11-05	₹5,408.06	income	Salary	Bonus credited
1,995	bcd	2025-11-05	₹4,477.86	income	Miscellaneous	Gift received
1,994	bcd	2025-11-05	₹11,894.63	income	Loan_Repayment	Bank interest credited

Transaction Insights

Total Amount

₹331,350.08

Expense Transactions

48

Income Transactions

52

Category Management

Category Usage (12 Categories)

Category	Transaction Count	Total Amount	Avg Amount	Unique Users	First Used	Last Used
Utilities	498	₹1,412,952.10	₹2,837.25	2	2023-01-03	2025-10-25
Salary	286	₹1,318,227.38	₹4,609.19	2	2023-01-15	2025-11-09
Miscellaneous	228	₹907,850.86	₹3,981.80	2	2023-01-01	2025-11-05
Healthcare	228	₹519,845.92	₹2,280.03	2	2023-01-04	2025-11-07
Groceries	170	₹307,599.70	₹1,809.41	2	2023-01-04	2025-11-05
Rent	112	₹387,577.06	₹3,460.51	2	2023-02-12	2025-10-24
Entertainment	110	₹231,648.54	₹2,105.90	2	2023-02-07	2025-10-25
Shopping	100	₹280,754.96	₹2,807.55	2	2023-01-02	2025-10-28
Transport	90	₹165,593.40	₹1,839.93	2	2023-01-05	2025-10-19
Uncategorized	64	₹243,305.16	₹3,801.64	2	2023-01-06	2025-11-01

Category Insights

Most Used Category

Utilities

Highest Total Value

₹1,412,952.10

Avg Tx per Category

166.7

System Tools

System Health Check

Check System Health

Database Size

0.29 MB

Transactions Count

2000

Users Count

4

Goals Count

2

System health check completed at 2025-11-16T13:44:00.356912

System Information

Backend API: Running on port 5000

Admin Dashboard: Running on port 8502

Database: SQLite (expense.db)

User App: Running on port 8501

19.Conclusion

BudgetWise AI represents a comprehensive solution to modern personal financial management challenges. By leveraging artificial intelligence for categorization, forecasting, and goal coaching, the system provides users with intelligent insights and actionable recommendations for better financial decision-making.

The architecture demonstrates a robust full-stack implementation with proper security measures, performance optimization, and scalability considerations. The modular design allows for maintainability and future extensibility while providing immediate value through its core financial management capabilities.

This documentation serves as a comprehensive reference for understanding the system's architecture, functionality, and implementation details, providing a solid foundation for deployment, maintenance, and potential future development.